

Lab Experiments 18-20

Data Handling & Visualization: Python Code & Output

Experiment 18

Age & Body Fat Analysis: Descriptive Statistics, Boxplots, Scatter Plot and Q-Q Plot

PYTHON

Code

```
"""
Q18. Hospital Age and Body Fat Data Analysis
-----
A hospital tested the age and body fat (%fat) data for 18 randomly
selected adults. This script:
    1. Calculates the mean, median and standard deviation of
       'age' and '%fat' using Pandas.
    2. Draws boxplots for 'age' and '%fat'.
    3. Draws a scatter plot and a Q-Q plot based on these two variables.
"""

import pandas as pd
import matplotlib.pyplot as plt
import scipy.stats as stats

# -----
# 1. Build the data set (as given in the question) into a DataFrame
# -----
age = [23, 23, 27, 27, 39, 41, 47, 49, 50,
       52, 54, 54, 56, 57, 58, 58, 60, 61]

fat = [9.5, 26.5, 7.8, 17.8, 31.4, 25.9, 27.4, 27.2, 31.2,
       34.6, 42.5, 28.8, 33.4, 30.2, 34.1, 32.9, 41.2, 35.7]

df = pd.DataFrame({"age": age, "%fat": fat})
print("Data set:")
print(df, "\n")

# -----
# 2. Mean, Median, Standard deviation
# -----
summary = pd.DataFrame({
    "Mean": df.mean(),
    "Median": df.median(),
    "Std Dev": df.std()          # sample standard deviation (ddof=1)
})
print("Summary statistics:")
print(summary, "\n")

# -----
# 3. Boxplots for age and %fat
# -----
fig, axes = plt.subplots(1, 2, figsize=(10, 5))
df.boxplot(column="age", ax=axes[0])
axes[0].set_title("Boxplot of Age")
axes[0].set_ylabel("Age")

df.boxplot(column="%fat", ax=axes[1])
axes[1].set_title("Boxplot of %Fat")
axes[1].set_ylabel("%Fat")

plt.tight_layout()
plt.savefig("boxplots_age_fat.png", dpi=150)
plt.close()
```

```

# -----
# 4. Scatter plot: age vs %fat
# -----
plt.figure(figsize=(6, 5))
plt.scatter(df["age"], df["%fat"], color="steelblue", edgecolor="black")
plt.title("Scatter Plot: Age vs %Fat")
plt.xlabel("Age")
plt.ylabel("%Fat")
plt.grid(True, linestyle="--", alpha=0.5)
plt.tight_layout()
plt.savefig("scatter_age_fat.png", dpi=150)
plt.close()

# -----
# 5. Q-Q plots (checking normality) for age and %fat
# -----
fig, axes = plt.subplots(1, 2, figsize=(10, 5))

stats.probplot(df["age"], dist="norm", plot=axes[0])
axes[0].set_title("Q-Q Plot of Age")

stats.probplot(df["%fat"], dist="norm", plot=axes[1])
axes[1].set_title("Q-Q Plot of %Fat")

plt.tight_layout()
plt.savefig("qqplots_age_fat.png", dpi=150)
plt.close()

print("Saved: boxplots_age_fat.png, scatter_age_fat.png, qqplots_age_fat.png")

```

Output

Data set:

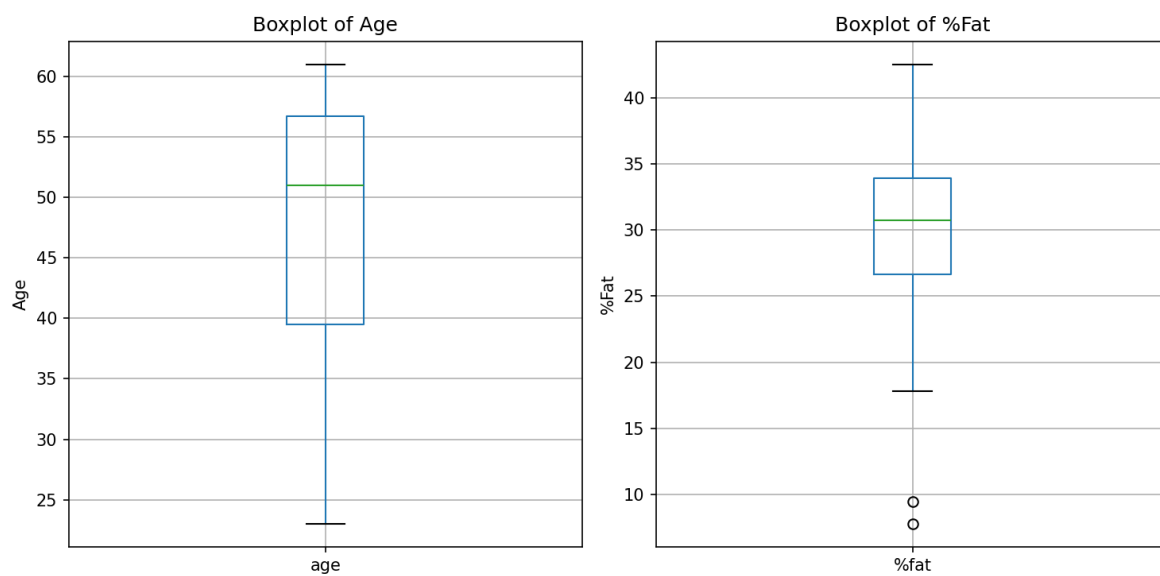
	age	%fat
0	23	9.5
1	23	26.5
2	27	7.8
3	27	17.8
4	39	31.4
5	41	25.9
6	47	27.4
7	49	27.2
8	50	31.2
9	52	34.6
10	54	42.5
11	54	28.8
12	56	33.4
13	57	30.2
14	58	34.1
15	58	32.9
16	60	41.2
17	61	35.7

Summary statistics:

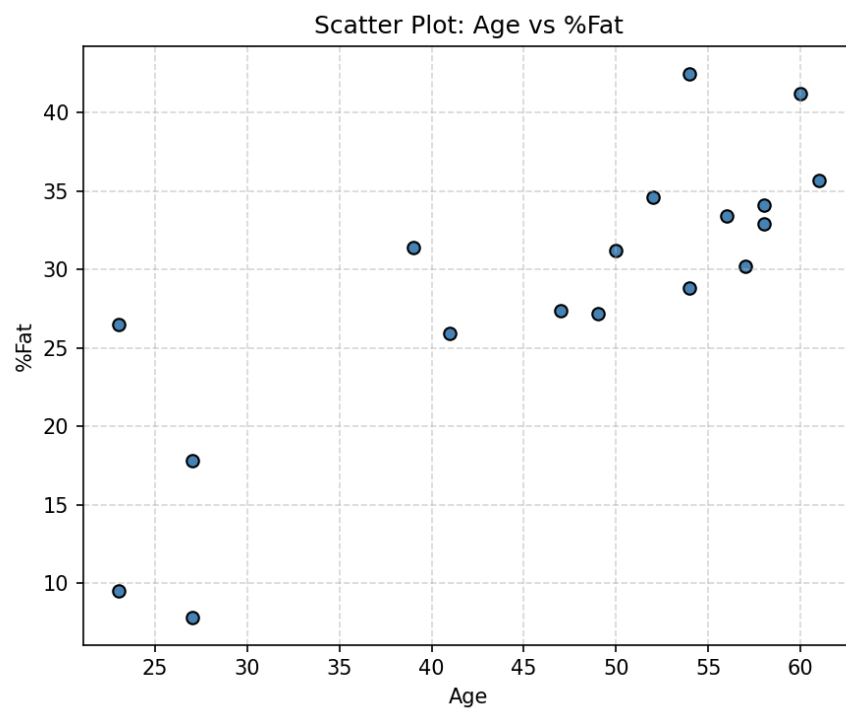
	Mean	Median	Std Dev
age	46.444444	51.0	13.218624
%fat	28.783333	30.7	9.254395

Saved: boxplots_age_fat.png, scatter_age_fat.png, qqplots_age_fat.png

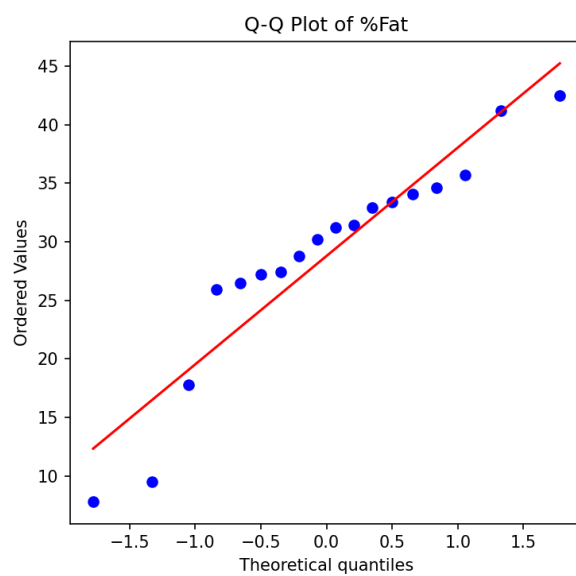
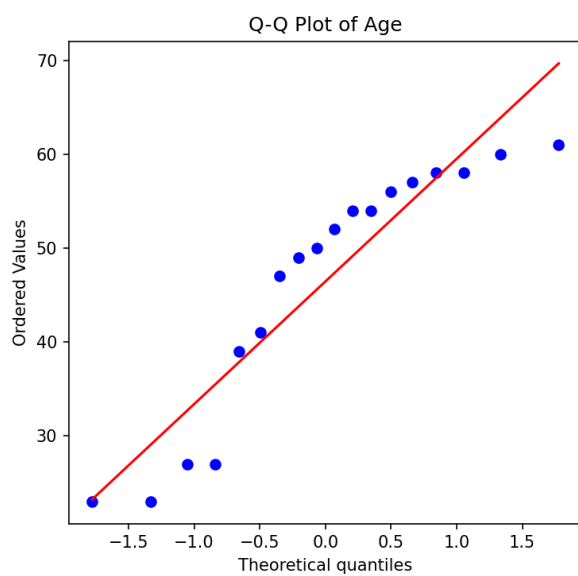
Boxplots of Age and %Fat



Scatter Plot: Age vs %Fat



Q-Q Plots of Age and %Fat



Experiment 19

Sales and Profit Analysis (Pandas)

PYTHON

Code

```
"""
Q19. Sales and Profit Analysis
-----
a) Load "sales_data.csv" into a Pandas DataFrame. Expected columns:
   'Date', 'Product', 'Quantity Sold', 'Unit Price'.
b) Create a new column 'Total Sales' = Quantity Sold * Unit Price.
c) Calculate the total sales for each product and the overall profit
   (assuming a 20% profit margin on each product). Display the top 5
   most profitable products.

NOTE: Place "sales_data.csv" in the same folder as this script before
running it. If the file is not found, a small sample file is
generated automatically so the script can still be demonstrated.
"""

import os
import pandas as pd

CSV_FILE = "sales_data.csv"
PROFIT_MARGIN = 0.20 # 20% profit margin

# -----
# Create a sample sales_data.csv if it does not already exist
# (so the script can be run/tested end-to-end).
# Remove this block once you have your real data file.
# -----
if not os.path.exists(CSV_FILE):
    sample = pd.DataFrame({
        "Date": ["2024-01-01", "2024-01-02", "2024-01-03", "2024-01-04",
                 "2024-01-05", "2024-01-06", "2024-01-07", "2024-01-08",
                 "2024-01-09", "2024-01-10"],
        "Product": ["Laptop", "Mouse", "Keyboard", "Monitor", "Laptop",
                    "Webcam", "Mouse", "Monitor", "Keyboard", "Webcam"],
        "Quantity Sold": [3, 10, 7, 4, 2, 6, 8, 5, 9, 12],
        "Unit Price": [750.00, 15.00, 25.00, 180.00, 750.00,
                       40.00, 15.00, 180.00, 25.00, 40.00]
    })
    sample.to_csv(CSV_FILE, index=False)
    print(f"'{CSV_FILE}' not found – a sample file was generated for demo purposes.\n")

# -----
# a) Load the CSV file into a DataFrame
# -----
df = pd.read_csv(CSV_FILE)
print("Loaded data:")
print(df, "\n")

# -----
# b) Create 'Total Sales' column
# -----
df["Total Sales"] = df["Quantity Sold"] * df["Unit Price"]
print("Data with 'Total Sales' column:")
print(df, "\n")

# -----
# c) Total sales per product & overall profit (20% margin)
# -----
product_sales = (
    df.groupby("Product")["Total Sales"]
    .sum()
```

```

        .reset_index()
        .rename(columns={"Total Sales": "Total Sales Amount"})
    )

# Profit per product = 20% of that product's total sales
product_sales["Profit"] = product_sales["Total Sales Amount"] * PROFIT_MARGIN

# Overall profit across all products
overall_profit = product_sales["Profit"].sum()

print("Total sales & profit by product:")
print(product_sales, "\n")

print(f"Overall profit (20% margin on all sales): {overall_profit:.2f}\n")

# -----
# Top 5 most profitable products
# -----
top5 = product_sales.sort_values("Profit", ascending=False).head(5)
print("Top 5 most profitable products:")
print(top5.to_string(index=False))

```

Output

Loaded data:

	Date	Product	Quantity Sold	Unit Price
0	2024-01-01	Laptop	3	750.0
1	2024-01-02	Mouse	10	15.0
2	2024-01-03	Keyboard	7	25.0
3	2024-01-04	Monitor	4	180.0
4	2024-01-05	Laptop	2	750.0
5	2024-01-06	Webcam	6	40.0
6	2024-01-07	Mouse	8	15.0
7	2024-01-08	Monitor	5	180.0
8	2024-01-09	Keyboard	9	25.0
9	2024-01-10	Webcam	12	40.0

Data with 'Total Sales' column:

	Date	Product	Quantity Sold	Unit Price	Total Sales
0	2024-01-01	Laptop	3	750.0	2250.0
1	2024-01-02	Mouse	10	15.0	150.0
2	2024-01-03	Keyboard	7	25.0	175.0
3	2024-01-04	Monitor	4	180.0	720.0
4	2024-01-05	Laptop	2	750.0	1500.0
5	2024-01-06	Webcam	6	40.0	240.0
6	2024-01-07	Mouse	8	15.0	120.0
7	2024-01-08	Monitor	5	180.0	900.0
8	2024-01-09	Keyboard	9	25.0	225.0
9	2024-01-10	Webcam	12	40.0	480.0

Total sales & profit by product:

	Product	Total Sales Amount	Profit
0	Keyboard	400.0	80.0
1	Laptop	3750.0	750.0
2	Monitor	1620.0	324.0
3	Mouse	270.0	54.0
4	Webcam	720.0	144.0

Overall profit (20% margin on all sales): 1352.00

Top 5 most profitable products:

	Product	Total Sales Amount	Profit
	Laptop	3750.0	750.0
	Monitor	1620.0	324.0
	Webcam	720.0	144.0
	Keyboard	400.0	80.0
	Mouse	270.0	54.0

Experiment 20

Customer Segmentation Based on Total Spending

PYTHON

Code

```
"""
Q20. Customer Segmentation
-----
a) Load "customer_data.csv" into a Pandas DataFrame. Expected columns:
   'Customer ID', 'Age', 'Gender', 'Total Spending'.
b) Segment customers into three groups based on Total Spending:
   "High Spenders", "Medium Spenders", "Low Spenders". Assign these
   segments to a new column in the DataFrame.
c) Calculate the average age of customers in each spending segment.

NOTE: Place "customer_data.csv" in the same folder as this script
before running it. If the file is not found, a small sample file is
generated automatically so the script can still be demonstrated.
"""

import os
import numpy as np
import pandas as pd

CSV_FILE = "customer_data.csv"

# -----
# Create a sample customer_data.csv if it does not already exist
# (so the script can be run/tested end-to-end).
# Remove this block once you have your real data file.
# -----
if not os.path.exists(CSV_FILE):
    rng = np.random.default_rng(42)
    n = 20
    sample = pd.DataFrame({
        "Customer ID": range(1, n + 1),
        "Age": rng.integers(18, 65, size=n),
        "Gender": rng.choice(["Male", "Female"], size=n),
        "Total Spending": rng.integers(50, 5000, size=n)
    })
    sample.to_csv(CSV_FILE, index=False)
    print(f"'{CSV_FILE}' not found – a sample file was generated for demo purposes.\n")

# -----
# a) Load the CSV file into a DataFrame
# -----
df = pd.read_csv(CSV_FILE)
print("Loaded data:")
print(df, "\n")

# -----
# b) Segment customers into three groups based on Total Spending
#     using tertiles (equal-sized groups). pd.qcut splits the data
#     into three bins with (roughly) equal numbers of customers.
# -----
df["Spending Segment"] = pd.qcut(
    df["Total Spending"],
    q=3,
    labels=["Low Spenders", "Medium Spenders", "High Spenders"]
)

print("Data with 'Spending Segment' column:")
print(df, "\n")

# -----
```

```
# c) Average age of customers in each spending segment
# -----
avg_age_by_segment = (
    df.groupby("Spending Segment", observed=True)["Age"]
      .mean()
      .reset_index()
      .rename(columns={"Age": "Average Age"})
)

print("Average age of customers by spending segment:")
print(avg_age_by_segment.to_string(index=False))
```

Output

Loaded data:

	Customer ID	Age	Gender	Total Spending
0	1	22	Female	867
1	2	54	Male	3802
2	3	48	Male	3517
3	4	38	Female	1804
4	5	38	Female	386
5	6	58	Female	4854
6	7	22	Male	2256
7	8	50	Female	4470
8	9	27	Female	3405
9	10	22	Male	3902
10	11	42	Male	3811
11	12	63	Male	1013
12	13	52	Male	1851
13	14	53	Female	2360
14	15	51	Female	2514
15	16	54	Male	266
16	17	42	Female	2755
17	18	24	Female	813
18	19	57	Male	3729
19	20	39	Female	3431

Data with 'Spending Segment' column:

	Customer ID	Age	Gender	Total Spending	Spending Segment
0	1	22	Female	867	Low Spenders
1	2	54	Male	3802	High Spenders
2	3	48	Male	3517	High Spenders
3	4	38	Female	1804	Low Spenders
4	5	38	Female	386	Low Spenders
5	6	58	Female	4854	High Spenders
6	7	22	Male	2256	Medium Spenders
7	8	50	Female	4470	High Spenders
8	9	27	Female	3405	Medium Spenders
9	10	22	Male	3902	High Spenders
10	11	42	Male	3811	High Spenders
11	12	63	Male	1013	Low Spenders
12	13	52	Male	1851	Low Spenders
13	14	53	Female	2360	Medium Spenders
14	15	51	Female	2514	Medium Spenders
15	16	54	Male	266	Low Spenders
16	17	42	Female	2755	Medium Spenders
17	18	24	Female	813	Low Spenders
18	19	57	Male	3729	High Spenders
19	20	39	Female	3431	Medium Spenders

Average age of customers by spending segment:

Spending Segment	Average Age
Low Spenders	41.571429
Medium Spenders	39.000000
High Spenders	47.285714